

HAT-P-54b

R-1585

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_190127 · created 2026-08-02T00:24:02+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458510.8201 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.024
Transit depth (Rp/Rs)^2	2.87 +/- 0.83 [%]
Orbital Inclination (inc)	84.86 +/- 1.13
Airmass coefficient 1 (a1)	0.975 +/- 0.023
Airmass coefficient 2 (a2)	0.024 +/- 0.018
Scatter in the residuals of the lightcurve fit is	2.38 %
Best Comparison Star	#2 - [355, 99]
Optimal Aperture	2.79
Optimal Annulus	12.54
Transit Duration (day)	0.0568 +/- 0.0092

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.169 ± 0.024 vs 0.1572 (deviation 0.48σ)
Duration fit vs published	0.0568 d vs 0.075 d (deviation 1.93σ)
Mid-time O–C	1.0 min
Depth SNR	6.9
Residual scatter	2.38 %

Light curve

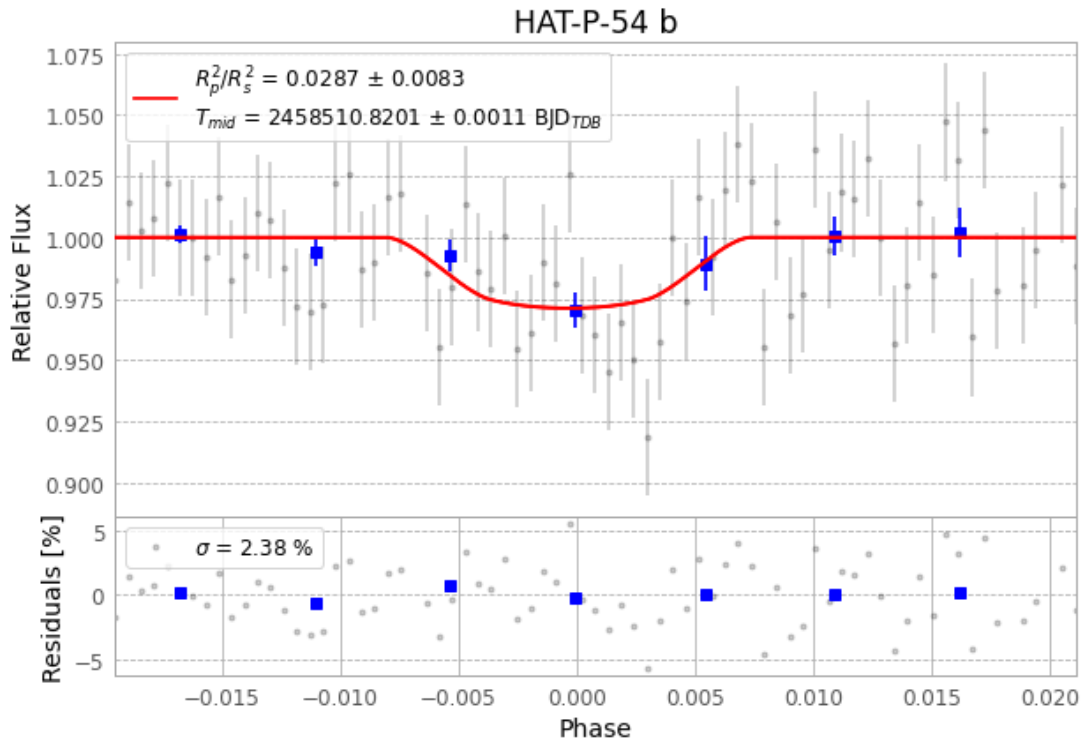


Figure 1 — FinalLightCurve_HAT-P-54 b_2019-01-26.png (SHA-256 72a6dfbe4e9065c9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [399, 210], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1c0ddcaaa958d9a2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), HATP-54190127054512.FITS → HATP-54190127092712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-190127.log (04afd364e864...), FinalLightCurve_HAT-P-54 b_2019-01-26.png (72a6dfbe4e90...), FinalLightCurve_HAT-P-54 b_2019-01-26.csv (c92cde74b311...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 00:24Z.